

8. String to Integer (atoi)

Medium Topics Companies

Implement the `myAtoi(string s)` function, which converts a string to a 32-bit signed integer.

The algorithm for `myAtoi(string s)` is as follows:

- Whitespace:** Ignore any leading whitespace (" ").
- Signedness:** Determine the sign by checking if the next character is '-' or '+', assuming positivity if neither present.
- Conversion:** Read the integer by skipping leading zeros until a non-digit character is encountered or the end of the string is reached. If no digits were read, then the result is 0.
- Rounding:** If the integer is out of the 32-bit signed integer range $[-2^{31}, 2^{31} - 1]$, then round the integer to remain in the range. Specifically, integers less than -2^{31} should be rounded to -2^{31} , and integers greater than $2^{31} - 1$ should be rounded to $2^{31} - 1$.

Return the integer as the final result.

Example 1:

Input: s = "42"

Output: 42

Explanation:

The underlined characters are what is read in and the caret is the current reader position.

Step 1: "42" (no characters read because there is no leading whitespace)

Step 2: "42" (no characters read because there is neither a '-' nor '+')

Step 3: "42" ("42" is read in)

Example 2:

6.5K 1K 141 Online

Java Auto

```
12 // 2. Check sign
13 int sign = 1;
14
15 if (i < n && s.charAt(i) == '-') {
16     sign = -1;
17     i++;
18 } else if (i < n && s.charAt(i) == '+') {
19     i++;
20 }
21
22 // 3. Convert digits
23 long result = 0;
24
25 while (i < n && Character.isDigit(s.charAt(i))) {
26
27     int digit = s.charAt(i) - '0';
28     result = result * 10 + digit;
29
30     // 4. Check 32-bit integer overflow
31     if (sign == 1 && result > Integer.MAX_VALUE) {
32         return Integer.MAX_VALUE;
33     }
34
35     if (sign == -1 && -result < Integer.MIN_VALUE) {
36         return Integer.MIN_VALUE;
37     }
38
39     i++;
40 }
41
42 return (int) (result * sign);
43 }
44 }
45
```

Saved

Ln 45, Col 1

Testcase Test Result

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Step 1: `"42"` (no characters read because there is no leading whitespace)

Step 2: `"42"` (no characters read because there is neither a `'-'` nor `'+'`)

Step 3: `"42"` ("`42`" is read in)

Example 2:

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3 Case 4 Case 5

Input

`s =
"42"`

Output

42

Expected

42

Contribute a testcase